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Math 208 Final Exam Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Let $A = \begin{bmatrix} 5 & -3 \\ 2 & 0 \end{bmatrix}$, $\Phi(t) = \begin{bmatrix} e^{2t} & 3e^{3t} \\ e^{2t} & 2e^{3t} \end{bmatrix}$. If $\Phi(t)$ is a fundamental matrix for the system $X' = AX$, then $e^{At} =$

(a) $\begin{bmatrix} -2e^{2t} + 3e^{3t} & 3e^{2t} - 3e^{3t} \\ -2e^{2t} + 2e^{3t} & 3e^{2t} - 2e^{3t} \end{bmatrix}$

(b) $\begin{bmatrix} 2e^{2t} - 3e^{3t} & -3e^{2t} + 3e^{3t} \\ 2e^{2t} - 2e^{3t} & -3e^{2t} + 2e^{3t} \end{bmatrix}$

(c) $\begin{bmatrix} -2e^{2t} + 9e^{3t} & e^{2t} - 3e^{3t} \\ -2e^{2t} + 6e^{3t} & e^{2t} - 2e^{3t} \end{bmatrix}$

(d) $\begin{bmatrix} 2e^{2t} + 3e^{3t} & 3e^{2t} + 3e^{3t} \\ 2e^{2t} + 2e^{3t} & 3e^{2t} + 2e^{3t} \end{bmatrix}$

(e) $\begin{bmatrix} e^{2t} + e^{3t} & e^{2t} - e^{3t} \\ e^{2t} - e^{3t} & e^{2t} + e^{3t} \end{bmatrix}$

2. The general solution of the differential equation

$$D(D-1)^2(D^2 + 4D + 5)y = 0$$

is

(a) $y = c_1 + c_2e^x + c_3e^{-x}\cos(2x) + c_4e^{-x}\sin(2x)$

(b) $y = c_1 + c_2e^{-x} + c_3xe^{-x} + c_4e^x\cos(2x) + c_5e^x\sin(2x)$

(c) $y = c_1 + c_2e^x + c_3e^{-2x}\cos x + c_4e^{-2x}\sin x$

(d) $y = c_1 + c_2e^x + c_3xe^x + c_4e^{-2x}\cos x + c_5e^{-2x}\sin x$

(e) $y = c_1 + c_2e^x + c_3xe^x + c_4e^x\cos(2x) + c_5e^x\sin(2x)$

3. Let $t = (4, 20, 23)$, $u = (1, 3, 2)$, $v = (2, 8, 7)$, $w = (1, 7, 9)$ be vectors in $\mathbb{R}^3$. If $t = au + bv + cw$, then $a + b + c =$

(a) 6

(b) 5

(c) 8

(d) -7

(e) 4

4. An appropriate form of a particular solution of the differential equation

$$y'' - y = 2e^x - 3e^{-x}$$

is

(a) $y_p = Ae^x + Be^{-x}$

(b) $y_p = Axe^x + Bxe^{-x}$

(c) $y_p = A + Be^x + Ce^{-x}$

(d) $y_p = Axe^x + Be^{-x}$

(e) $y_p = Ae^x + Bxe^x$

5. Let $A = \begin{bmatrix} 3 & 6 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. A basis for the eigenspace of A associated with the eigenvalue $\lambda = 1$ of A is

(a) $\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \right\}$

(b) $\left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$

(c) $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right\}$

(d) $\left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \right\}$

(e) $\left\{ \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} \right\}$

6. Find a general solution of the system $X' = \begin{bmatrix} 4 & 2 \\ -3 & -1 \end{bmatrix} X$.

(a) $X = c_1 \begin{bmatrix} e^t \\ -e^t \end{bmatrix} + c_2 \begin{bmatrix} 2e^{2t} \\ -e^{2t} \end{bmatrix}$

(b) $X = c_1 \begin{bmatrix} 2e^t \\ -3e^t \end{bmatrix} + c_2 \begin{bmatrix} e^{2t} \\ -e^{2t} \end{bmatrix}$

(c) $X = c_1 \begin{bmatrix} 2e^{-t} \\ e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^{-2t} \\ -e^{-2t} \end{bmatrix}$

(d) $X = c_1 \begin{bmatrix} -2e^t \\ 3e^t \end{bmatrix} + c_2 \begin{bmatrix} 2e^{2t} \\ 3e^{2t} \end{bmatrix}$

(e) $X = c_1 \begin{bmatrix} 2e^{-t} \\ -3e^{-t} \end{bmatrix} + c_2 \begin{bmatrix} e^{-2t} \\ -e^{-2t} \end{bmatrix}$

7. If $y(x)$ is the solution of the initial value problem

$$y'' + y = 1, \quad y(0) = 2, \quad y'(0) = 1,$$

then $y\left(\frac{\pi}{4}\right) =$

(a) $3 + 2\sqrt{2}$

(b) $1 + \sqrt{2}$

(c) $2 + 3\sqrt{2}$

(d) $1 - 3\sqrt{2}$

(e) $2 + \sqrt{2}$

8. If the general solution of the system

$$X' = \begin{bmatrix} 6 & -4 & 6 \\ 1 & 6 & 1 \\ 1 & 6 & 1 \end{bmatrix} X$$

is $X = c_1 \begin{bmatrix} 1 \\ 1 \\ a \end{bmatrix} e^{8t} + c_2 \begin{bmatrix} b \\ -1 \\ -1 \end{bmatrix} e^{5t} + c_3 \begin{bmatrix} 1 \\ c \\ -1 \end{bmatrix}$, then $a + b + c =$

(a) 1

(b) 3

(c) 0

(d) 2

(e) 4

9. If $y(x)$ is the solution of the initial value problem

$$y' + e^x y^2 = 0, \qquad y(0) = \frac{1}{2},$$

then $y(\ln 2) =$

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) $\frac{1}{4}$

(d) $\frac{3}{4}$

(e) $\frac{2}{3}$

10. If $\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} - \sum_{n=1}^{\infty} c_n x^{n-1} = 0$, then $\dfrac{c_{26}}{9c_{27}} =$

(a) 81

(b) 48

(c) 64

(d) 78

(e) 99

11. Solve the homogeneous differential equation

$$2xy\frac{dy}{dx} = 4x^2 + 3y^2.$$

(a) $y^3 - x^3 = x^2 + k$

(b) $y^2 + 4x^2 = kx^3$

(c) $y^2 - 2x^2 = kx^3$

(d) $3y^2 + x^3 = kx^4$

(e) $-5y^2 + x^2 = kx^3$

12. Let A be a 10×15 matrix. If the rank of A is 7, then the dimension of the solution space of the system $AX = 0$ is equal to

(a) 8

(b) 15

(c) 7

(d) 3

(e) 10

13. The matrix

$$A = \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{bmatrix}$$

has a defective eigenvalue $\lambda = -1$ of defect 2. Choosing $v_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ such that $(A+I)^3 v_3 = 0$ and $(A+I)^2 v_3 \neq 0$, then the general solution of the system $X' = AX$ is

(a) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 1 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} t^2/2 \\ 1+2t \\ t \end{bmatrix} \right) e^{-t}$

(b) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 2 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} t^2/2 \\ -t \\ 1+t \end{bmatrix} \right) e^{-t}$

(c) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 2 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} t^2/2 \\ 1+2t \\ t \end{bmatrix} \right) e^{-t}$

(d) $X = \left(c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -t \\ 2 \\ -1 \end{bmatrix} t + c_3 \begin{bmatrix} t^2 \\ t \\ t \end{bmatrix} \right) e^{-t}$

(e) $X = \left(c_1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} t \\ 0 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} t^2 \\ 2t \\ 2 \end{bmatrix} \right) e^{-t}$

14. A possible fundamental matrix for the system $X' = \begin{bmatrix} 6 & 2 \\ 5 & 3 \end{bmatrix} X$ is

(a) $\Phi(t) = \begin{bmatrix} 2e^t & 2e^{8t} \\ -5e^t & 2e^{8t} \end{bmatrix}$

(b) $\Phi(t) = \begin{bmatrix} 4e^t & 2e^{8t} \\ -10e^t & -3e^{8t} \end{bmatrix}$

(c) $\Phi(t) = \begin{bmatrix} -2e^t & -e^{8t} \\ 5e^t & e^{8t} \end{bmatrix}$

(d) $\Phi(t) = \begin{bmatrix} -e^t & 6e^{8t} \\ 5e^t & 6e^{8t} \end{bmatrix}$

(e) $\Phi(t) = \begin{bmatrix} e^t & -2e^{8t} \\ e^t & 5e^{8t} \end{bmatrix}$

15. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. If P is a diagonalizing matrix such that $P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$, then

(a) $P = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

(b) $P = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}$

(c) $P = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$

(d) $P = \begin{bmatrix} -1 & -1 \\ 1 & 0 \end{bmatrix}$

(e) $P = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$

16. An eigenvector associated with the eigenvalue $\lambda = 9i$ of the matrix

$$A = \begin{bmatrix} 0 & -3 \\ 27 & 0 \end{bmatrix}$$

is

(a) $\begin{bmatrix} i \\ 1 \end{bmatrix}$

(b) $\begin{bmatrix} 2i \\ -1 \end{bmatrix}$

(c) $\begin{bmatrix} i \\ 3 \end{bmatrix}$

(d) $\begin{bmatrix} -i \\ 3 \end{bmatrix}$

(e) $\begin{bmatrix} -3i \\ -6 \end{bmatrix}$

17. If $X = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{3t} + c_2 \begin{bmatrix} 1 \\ 4 \end{bmatrix} e^{-3t}$ is the solution of the initial value problem

$$X' = \begin{bmatrix} 5 & -2 \\ 8 & -5 \end{bmatrix} X, \quad X(0) = \begin{bmatrix} 5 \\ -4 \end{bmatrix},$$

then $c_1 - c_2 =$

(a) 10

(b) 12

(c) 13

(d) 9

(e) 11

18. The guaranteed radius of convergence of the power series solution of

$$(x^2+4)y'' + 4xy' + 2y = 0$$

about the ordinary point $x = 0$ is

(a) $\sqrt{2}$

(b) ∞

(c) 4

(d) 2

(e) 0

19. If the rank of the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & k \\ 1 & 2 & 2 & 1 \\ 1 & 2 & k & 1 \end{bmatrix}$$

is equal to 2, then $k =$

(a) -1

(b) -2

(c) 1

(d) 2

(e) 0

20. Let

$$F(t) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \qquad e^{At} = \begin{bmatrix} \cos(2t) & -\sin(2t) \\ \sin(2t) & \cos(2t) \end{bmatrix}.$$

A particular solution for the system $X' = AX + F(t)$ is

(a) $X_p = \begin{bmatrix} 2\cos(2t) \\ -2\sin(2t) \end{bmatrix}$

(b) $X_p = \begin{bmatrix} -2 \\ 0 \end{bmatrix}$

(c) $X_p = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

(d) $X_p = \begin{bmatrix} \sin(2t) \\ \cos(2t) \end{bmatrix}$

(e) $X_p = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

21. Using variation of parameters, the differential equation

$$y'' + 4y = 4\sec(2x)$$

has a particular solution $y_p = u_1y_1 + u_2y_2$. Then $u_1 + u_2 =$

(a) $2x + \ln|\sec(2x) + \tan(2x)|$

(b) $2x + \ln|\cos(2x)|$

(c) $2x - \ln|\sin(2x)|$

(d) $2x + \ln|\csc(2x) + \cot(2x)|$

(e) $4x - \ln|\cos(2x)|$

22. A 2×2 matrix A has an eigenvector $\begin{bmatrix} 1+2i \\ 2 \end{bmatrix}$ associated with the eigenvalue $\lambda = 4i$ of A. Then the general solution of the system $X' = AX$ is

(a) $X = c_1 \begin{bmatrix} -\cos(4t) + 2\sin(4t) \\ 2\cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(4t) + \sin(4t) \\ 2\sin(4t) \end{bmatrix}$

(b) $X = c_1 \begin{bmatrix} 2\cos(4t) - \sin(4t) \\ 3\cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(4t) - \sin(4t) \\ 3\sin(4t) \end{bmatrix}$

(c) $X = c_1 \begin{bmatrix} \cos(4t) + \sin(4t) \\ \cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} \cos(4t) + 2\sin(4t) \\ -2\sin(4t) \end{bmatrix}$

(d) $X = c_1 \begin{bmatrix} \cos(4t) + 3\sin(4t) \\ \cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} \cos(4t) + \sin(4t) \\ \sin(4t) \end{bmatrix}$

(e) $X = c_1 \begin{bmatrix} \cos(4t) - 2\sin(4t) \\ 2\cos(4t) \end{bmatrix} + c_2 \begin{bmatrix} 2\cos(4t) + \sin(4t) \\ 2\sin(4t) \end{bmatrix}$

23. In a certain culture of bacteria, the number of bacteria doubled after 4 hours. How long did it take for the population to triple. (Assume the rate of change of population is proportional to the population present at time t).

(a) $\dfrac{1}{\ln 2}$

(b) $\dfrac{\ln 27}{\ln 2}$

(c) $\dfrac{\ln 9}{\ln 2}$

(d) $\dfrac{\ln 81}{\ln 2}$

(e) $\dfrac{\ln 3}{\ln 2}$

24. The matrix

$$A = \begin{bmatrix} 2 & 0 & 2 \\ 1 & 1 & 1 \\ \frac{1}{2} & 0 & 2 \end{bmatrix}$$

has

(a) one eigenvalue of multiplicity 1 and one eigenvalue of multiplicity 2

(b) no real eigenvalues

(c) one eigenvalue of multiplicity 3

(d) one real and one pair of nonreal complex eigenvalues

(e) three distinct real eigenvalues

25. The sum of the roots of the indicial equation at $x = 0$ for $x^2y'' + 7xy' - 7y = 0$ is

(a) 7

(b) -1

(c) 0

(d) 4

(e) -6

26. The dimension of the subspace

$$W = \{(x_1, x_2, x_3, x_4, x_5) : x_1 - x_2 = x_3 + 2x_4 - 3x_5\}$$

of $\mathbb{R}^5$ is equal to

(a) 3

(b) 2

(c) 5

(d) 1

(e) 4

27. Let $y = \sum_{k=0}^{\infty} c_k x^k$ be a power series solution of $y'' - x^2y = 0$ about $x = 0$. Then the coefficients satisfy

(a) $c_{k+2} = \dfrac{c_{k-2}}{k(k-1)}$, $k \geq 2$

(b) $c_{k+2} = \dfrac{c_{k-1}}{k(k+1)}$, $k \geq 1$

(c) $c_{k+2} = \dfrac{c_k}{(k+1)(k+2)}$, $k \geq 0$

(d) $c_{k+2} = \dfrac{c_{k-2}}{(k+1)(k+2)}$, $k \geq 2$

(e) $c_{k+3} = \dfrac{c_{k+1}}{k(k+1)}$, $k \geq 1$

28. If $v = \begin{bmatrix} a \\ 1 \\ b \end{bmatrix}$ is an eigenvector of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$$

associated with the eigenvalue $\lambda = 3$, then $4a + 2b =$

(a) 6

(b) -1

(c) -3

(d) 1

(e) 3

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$e^{At} = \Phi(t)\Phi(0)^{-1}$ with $\Phi(0)^{-1} = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$.
2	**(d)**	Roots $0, 1, 1, -2 \pm i$: $c_1 + c_2 e^x + c_3 x e^x + e^{-2x}(c_4 \cos x + c_5 \sin x)$.
3	**(a)**	Row reduce: $c = 3$, $b = -2$, $a = 5$, so $a + b + c = 6$.
4	**(b)**	$y_c = c_1 e^x + c_2 e^{-x}$, both terms repeat, so $y_p = Axe^x + Bxe^{-x}$.
5	**(c)**	$A - I \to v_1 + 3v_2 - v_3 = 0$: basis $(1, 0, 1)$, $(-3, 1, 0)$.
6	**(b)**	$\lambda = 1, 2$ with eigenvectors $(2, -3)$, $(1, -1)$: $c_1(2, -3)e^t + c_2(1, -1)e^{2t}$.
7	**(b)**	$y = \cos x + \sin x + 1$, so $y(\pi/4) = 1 + \sqrt{2}$.
8	**(b)**	$Av = \lambda v$ for $\lambda = 8, 5, 0$: $a = 1$, $b = 2$, $c = 0$, sum 3.
9	**(a)**	Separable: $y = \frac{1}{e^x + 1}$, so $y(\ln 2) = \frac{1}{3}$.
10	**(d)**	$(k + 2)(k + 1)c_{k+2} = c_{k+1}$; $k = 25$: $\frac{c_{26}}{c_{27}} = 702$, so 78.
11	**(b)**	$v = y/x$: $xv' = \frac{v^2+4}{2v}$, $v^2 + 4 = kx$, so $y^2 + 4x^2 = kx^3$.
12	**(a)**	Dimension $= n - \text{rank} = 15 - 7 = 8$.
13	**(c)**	Chain $v_3 \to v_2 = (0, 2, 1) \to v_1 = (1, 0, 0)$, then X_1, X_2, X_3 with e^{-t}.
14	**(a)**	$\lambda = 1, 8$, eigenvectors $(2, -5)$, $(2, 2)$ as columns times $e^{\lambda t}$.

Q	Ans	Brief explanation
15	**(a)**	Eigenvectors $(-1,1)$ for 1 and $(1,1)$ for 3 as columns of P.
16	**(c)**	$-9iv_1 - 3v_2 = 0 \Rightarrow v_2 = -3iv_1$; $v_1 = i$ gives $(i, 3)$.
17	**(e)**	$c_1 + c_2 = 5$, $c_1 + 4c_2 = -4$: $c_1 = 8$, $c_2 = -3$, so $c_1 - c_2 = 11$.
18	**(d)**	Singular points $x = \pm 2i$ are at distance 2 from 0, so $R = 2$.
19	**(d)**	Row reduce to last row $(0, 0, k-2, 0)$; rank 2 iff $k = 2$.
20	**(e)**	$X_p = e^{At} \int e^{-At} F\, dt = e^{At}(\sin 2t, \cos 2t)^T = (0,1)^T$.
21	**(b)**	$W = 2$, $u_1' = -2\tan 2x$, $u_2' = 2$: $u_1 + u_2 = 2x + \ln\lvert\cos 2x\rvert$.
22	**(e)**	Re and Im of $(1+2i, 2)^T e^{4it}$ give the two solutions.
23	**(d)**	$k = \frac{\ln 2}{4}$, $e^{kt} = 3$: $t = \frac{4\ln 3}{\ln 2} = \frac{\ln 81}{\ln 2}$.
24	**(a)**	$\det(A - \lambda I) = (1-\lambda)^2(3-\lambda)$: $\lambda = 1$ (twice), 3 (once).
25	**(e)**	$r(r-1) + 7r - 7 = r^2 + 6r - 7 = 0$: roots $1, -7$, sum -6.
26	**(e)**	One equation in 5 unknowns: $5 - 1 = 4$ free variables, $\dim W = 4$.
27	**(d)**	Shift indices: $(k+2)(k+1)c_{k+2} = c_{k-2}$ for $k \geq 2$.
28	**(e)**	$Av = 3v$: $b = a$, $a + b = 1$, so $a = b = \frac{1}{2}$ and $4a + 2b = 3$.